

Linear Algebra — Homework 4

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25I-3207

SE-3A

Solutions:

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Question 1.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, W = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 : \begin{array}{l} x_1 + x_2 - x_3 = 0, \\ 2x_1 + 2x_2 - 2x_3 = 0 \end{array} \right\}$$

i- W is a subspace of $\mathbb{R}^4$ as $0 \in W$ given

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 2 & 2 & -2 & 0 \end{bmatrix}$$

• Closed under addition as well as scalar multiplication.

ii- $R_2' = 2R_1 - R_2$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Independent row is $x_1 + x_2 - x_3 = 0$

$$x_1 + x_2 = x_3$$

$$x_1 = x_1$$

$$x_2 = x_2$$

$$x_4 = x_4$$

$$\text{Hence } X = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} x_4$$

$$\text{Basis} = \left\{ (1, 0, 1, 0), (0, 1, 1, 0), (0, 0, 0, 1) \right\}$$

iii -

Dimension is 3 as 4×20 , and we got 3 basis only.
 $\dim(W) = 3$

iv -

The dimension of W is found by n -rank of matrix.

Rank counts independent rows. Since rank is 1. So $4-1=3$.

Question No 21.

$$V = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a+d=0, b=c \right\}$$

i -

Yes V is a subspace as it satisfies the three axioms,
 V has '0' vector, addition is possible & scalar multiplication
also results in space $\in \mathbb{R}^2$.

ii -

$$d = -a, c = b$$

a & b are free variables.

$$\text{So } \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ b & -a \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Basis : } \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$$

iii-

 $\dim(V) = 2$, same concept as in earlier question.

2

iv-

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \right\}$$

 $\dim(V) = 2$, but S has 3 vectors.

• Moreover, the third vector $= 2 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

So it's a span of already present basis.

• So ' S ' is not the basis of ' V '.

Question No 3.

$$W = \{ p(x) \in P_4 : p(1) = 0, p(-1) = 0 \}$$

i- Since $p(x) \in P_4$, it's a polynomial of degree 4.

Let $Q(p) = (p(1) + p(-1)) = 0 \in P_4$.

$cQ(p) = (cp(1) + cp(-1)) = c(p(1) + p(-1)) \in P_4$

• Hence W is subspace of P_4 .

ii-

$p(x) = 0$ at $p(1)$ & $p(-1)$

So $(x-1)(x+1) = x^2 - 1$

Degree of $x^2 - 1 = 2$, so only 3 basis are involved such

that $c(x^2 - 1)$, $c(x^3 - x)$ & $c(x^4 - x^2)$

So basis = $\{ (x^2 - 1), (x^3 - x), (x^4 - x^2) \}$.

iii-

$$\dim(W) = 3$$

iv-

General form of $p(x)$ in $W = (x-1)(x+1)q(x)$

where $q(x) = cx^2 + cx + c$ where c can vary as $\in \mathbb{R}$.

v- $x^4 - x^2 \in W$

$p(1) = 0$, $p(-1) = 0$, hence $\in W$.

Question No 4:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 4 \\ 3 \\ 7 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \end{bmatrix}, v_4 = \begin{bmatrix} 3 \\ 6 \\ 4 \\ 10 \end{bmatrix}$$

$$W = \text{span} \{v_1, v_2, v_3, v_4\}$$

i-

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 2 & 6 \\ 1 & 3 & 2 & 4 \\ 3 & 7 & 4 & 10 \end{bmatrix}$$

By row operations;

$$R_2' = 2R_1 - R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 1 & 3 & 2 & 4 \\ 3 & 7 & 4 & 10 \end{bmatrix}$$

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$$R_4' = R_4 - 2R_1 \rightarrow R_4, \quad R_3' = R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 3 & 2 & 4 \end{bmatrix}$$

$$R_4' = R_4 - R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} R_4' &= R_4 - R_3 \\ R_2 &\rightleftharpoons R_3 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1' = 2R_2 - R_1$$

$$\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

v_1 & v_2 are only independent variables.

Basis = $\{v_1, v_2\}$.

ii -

$$\dim(W) = 2$$

iii -

$$v_4 = v_1 + v_2$$

Hence it's just a linear combination and does not introduce a new direction.

iv-

Linear dependence of other vectors on the basis allow us to only store the instructions of linear combination and the basis instead of storing all vector spaces and all vectors in each space.

Hence data redundancy is avoided to optimize computational costs and memory usage.

Question No 5.

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

For v_3 & v_4 , we can choose vectors as

$$v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \& \quad v_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Testing:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} R_3' = R_3 - R_1 \\ R_4' = R_4 - R_1 \end{matrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3' = R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad I_4 \text{ (Identity matrix)}^{4 \times 4}$$

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- Rank = 4 as 4 columns are linearly independent.

4

$$v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

- Basis for $\mathbb{R}^4 = \{(1, 0, 1, 1), (0, 1, 1, 0), (0, 0, 1, 0), (0, 0, 0, 1)\}$.

Question No 6:

$$A = \begin{bmatrix} 1 & -1 & 2 & 0 & 3 \\ -2 & 2 & -4 & 1 & -5 \\ 0 & 0 & 0 & 2 & 2 \\ 3 & -3 & 6 & -1 & 8 \end{bmatrix}$$

i-

$$R_2' = 2R_1 + R_2$$

$$R_3' = R_3 - 3R_1$$

$$A = \begin{bmatrix} 1 & -1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & -1 & -1 \end{bmatrix}$$

$$R_1' = R_2 + R_1$$

$$R_3' = R_3 - 2R_2$$

$$A = \begin{bmatrix} 1 & -1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

rank of $A = 2$

ii-

$$\text{Basis of } \text{Col}(A) = \left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ -1 \end{bmatrix} \right\}$$

iii-

$$\text{Basis of } \text{Row}(A) = \{(1, -1, 2, 0, 3), (0, 0, 0, 1, 1)\}$$

iv-

Basis of $\text{Null}(A)$ is

$$x_1 - x_2 + 2x_3 + 3x_5 = 0$$

$$x_4 + x_5 = 0$$

$$\text{So } x_1 = x_2 - 2x_3 - 3x_5$$

$$x_4 = -x_5$$

where x_2, x_3 & x_5 are free variables.

$$x = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} -3 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} x_5$$

v-

Basis of left nullspace $\text{Null}(A^T)$

$$= \begin{bmatrix} -4 \\ -2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

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vi-

$$\dim(\text{Col } A) + \dim(\text{Nul } A) = 2 + 3 = 5 = n$$

$$\dim(\text{Row } A) + \dim(\text{Nul } A^T) = 2 + 2 = 4 = m.$$

Python Script:

```
import sympy as sp

A = sp.Matrix([
    [ 1, -1,  2,  0,  3],
    [-2,  2, -4,  1, -5],
    [ 0,  0,  0,  2,  2],
    [ 3, -3,  6, -1,  8]
])

print("Matrix A:")
sp.pprint(A)
print()

rref_matrix, pivot_columns = A.rref()
print("RREF of A:")
sp.pprint(rref_matrix)
print("Pivot columns (0-indexed):", pivot_columns)

rank_A = A.rank()
print(f"rank(A) = {rank_A}")
print()

col_space = A.columnspace()
print("Basis for Col(A):")
for vec in col_space:
    sp.pprint(vec.T)
print()

row_space = A.rowspace()
print("Basis for Row(A):")
for vec in row_space:
    sp.pprint(vec)
print()

null_space = A.nullspace()
print("Basis for Nul(A):")
for vec in null_space:
    sp.pprint(vec.T)
print()
```

```

left_null_space = A.T.nullspace()
print("Basis for Nul( $A^T$ ) (left null space):")
for vec in left_null_space:
    sp.pprint(vec.T)
print()

m, n = A.shape
r = rank_A
dim_col = len(col_space)
dim_row = len(row_space)
dim_null = len(null_space)
dim_inull = len(left_null_space)

print("--- Fundamental Theorem check ---")
print(f"n (columns of A) = {n}, m (rows of A) = {m}, rank(A) = {r}")
print(f"dim(Col A) + dim(Nul A) = {dim_col} + {dim_null} = {dim_col + dim_null} (should equal n = {n})")
print(f"dim(Row A) + dim(Nul  $A^T$ ) = {dim_row} + {dim_inull} = {dim_row + dim_inull} (should equal m = {m})")

assert dim_col + dim_null == n, "Rank-Nullity theorem failed for Col/Nul(A)!"
assert dim_row + dim_inull == m, "Rank-Nullity theorem failed for Row/Nul( $A^T$ )!"
print("Both identities verified. ✓")

print()
print("--- Verification ---")
for v in null_space:
    assert A * v == sp.zeros(m, 1), "A*v != 0 for a claimed null space vector!"
print("Confirmed: A @ v = 0 for every basis vector of Nul(A).")

for y in left_null_space:
    assert A.T * y == sp.zeros(n, 1), " $A^T$ *y != 0 for a claimed left-null vector!"
print("Confirmed: A.T @ y = 0 for every basis vector of Nul( $A^T$ ).")

```

Screenshot:

```
muhammad-awais@ubuntu: ~  
muhammad-awais@ubuntu: ~/Work/University-Assignments/Sem3/LA/HW4  
muhammad-awais@ubuntu: ~ -- /opt/google/chrome/chrome  
(.venv) muhammad-awais@ubuntu:~/Work/University-Assignments/Sem3/LA/HW4$ python3 verify.py  
Matrix A:  
[ 1 -1 2 0 3  
 -2 2 -4 1 -5  
 0 0 0 2 2  
 3 -3 6 -1 0]  
RREF of A:  
[ 1 -1 2 0 3  
 0 0 0 1 1  
 0 0 0 0 0  
 0 0 0 0 0  
 0 0 0 0 0]  
Pivot columns (0-indexed): (0, 3)  
rank(A) = 2  
Basis for Col(A):  
[ 1 -2 0 3  
 0 1 2 -1]  
Basis for Row(A):  
[ 1 -1 2 0 3  
 0 0 0 1 1]  
Basis for Nul(A):  
[ 1 2 0 0 0  
 -2 0 1 0 0  
 -3 0 0 -1 1]  
Basis for Nul(A^T) (left null space):  
[ -4 -2 1 0  
 -1 1 0 1]  
... Fundamental Theorem check ...  
n (columns of A) = 5, m (rows of A) = 4, rank(A) = 2  
dim(Col A) + dim(Nul A) = 2 + 3 = 5 (should equal n = 5)  
dim(Row A) + dim(Nul A^T) = 2 + 2 = 4 (should equal m = 4)  
Both identities verified. ✓  
... Verification ...  
Confirmed: A @ v = 0 for every basis vector of Nul(A).  
Confirmed: A.T @ y = 0 for every basis vector of Nul(A^T).  
(.venv) muhammad-awais@ubuntu:~/Work/University-Assignments/Sem3/LA/HW4$
```